

Dinesh Pandikona

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EDUCATION

M.S in Computer Science, Northeastern University <i>Machine Learning, Foundations of AI, Algorithms, Operating Systems</i>	May 2025 <i>Boston</i>
B.S in Computer Science, VNR VJIET <i>Web Development, Distributed Systems, Java Development, Math (Calculus)</i>	May 2022 <i>Hyderabad</i>

WORK EXPERIENCE

Software Engineer Intern Bose Professional, United States • Designed a distributed audio state-sync service using Memberlist for peer discovery and gossip updates, implementing CRDT-based conflict resolution in Go , to maintain consistency under intermittent networks. • Architected low-latency GraphQL APIs with optimized resolver pipelines, in-memory caching layers, and batched queries, reducing real-time audio sync latency by 25% for 100+ concurrent devices . • Built a rules-driven event engine in Go , using concurrency-safe channels and worker pools to trigger automated device actions based on real-time DSP parameter changes, improving response times for system-wide updates.	July 2024 - Dec 2024
Software Analyst Deloitte, India • Developed a real-time collaborative expense tracking module in React + TypeScript with PostgreSQL row-level locking and WebSocket-based state synchronization, enabling multi-user editing without data conflicts. • Engineered Java-based automation tools with multi-threaded job execution and JDBC batching , integrating SQL stored procedures and indexed queries to cut average runtime by 50% on datasets exceeding 50M records . • Implemented a data processing pipeline using parameterized workflows, enabling users to execute complex tasks with minimal technical knowledge while ensuring scalability and security.	Feb 2022 – Aug 2023
Machine Learning Engineer Intern Blaze Automation, India • Built an anomaly detection pipeline in Python using Scikit-learn and Pandas , achieving 86% accuracy on streaming telemetry from devices, preprocessing optimized via vectorized operations for 3× faster inference. • Designed a modular ML serving framework with configurable feature extractors and model endpoints, supporting real-time predictions for elderly-care monitoring systems deployed in 50+ homes . • Created and evaluated a lightweight time-series feature extraction library for embedded edge devices, balancing memory footprint with predictive accuracy; work resulted in a peer-reviewed research paper .	Feb 2021 – June 2021

PROJECTS

Distributed Rate Limiter Rust, Redis, Open-Source • Developed a scalable distributed rate limiter in Rust supporting multiple algorithms, leveraging Redis for fast state storage, achieving 10K+ RPS with sub-millisecond latency. • Implemented various resilience mechanisms including circuit breakers , health checks , and fallback strategies to ensure high availability and fault tolerance, achieving 99.99% uptime in simulated distributed failures. • Designed a comprehensive testing infrastructure with extensive unit and integration tests (100+ test cases) that validates algorithm accuracy, storage reliability, and resilience mechanisms under various failure scenarios	April 2025
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TECHNICAL SKILLS

Languages:	Python, Golang, TypeScript, Rust
Data/Analytics:	NumPy, Pandas, Apache Druid, RedShift basics
Distributed Systems/Streaming:	gRPC, GraphQL, REST, Redis, PostgreSQL, Kafka, HLS/RTMP basics
Frameworks:	Next.js (React.js), Node.js, PyTorch, Scikit-learn
Infrastructure/DevOps:	AWS (EC2, S3, RDS, Lambda), Docker, Kubernetes, Terraform, Prometheus
Tools:	Git, Jira (Agile), Jenkins, CI/CD pipelines